

MTI Probability and Statistics
(Lecture 6)

January 13, 2026

Probability Distribution of a Random Variable: So far we have introduced the concept of a random variable and noted that the concept of probability on the sample space was not used in this definition. In practice, however, random variables are of interest only when they are defined on a probability space. Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let X be a random variable defined on it.

Theorem 1: The random variable X defined on probability space $(\Omega, \mathcal{F}, P)$ induces a probability space $(\mathcal{R}, \mathcal{B}, Q)$ by means of the correspondence

$$Q(B) = P\{X^{-1}(B)\} = P\{\omega : X(\omega) \in B\} \quad \forall B \in \mathcal{B}. \quad (1)$$

We write $Q = P(X^{-1})$ and call Q the (probability) distribution of X .

Proof: Clearly, $Q(B) \geq 0 \quad \forall B \in \mathcal{B}$, and also $Q(\mathcal{R}) = P\{X \in \mathcal{R}\} = P(\Omega) = 1$. Let $B_i \in \mathcal{B}$, $i = 1, 2, \dots$, with $B_i \cap B_j = \emptyset$, $i \neq j$. Since the inverse image of a disjoint union of Borel sets is the disjoint union of their inverse images, we have

$$\begin{aligned} Q\left(\bigcup_{i=1}^{\infty} B_i\right) &= P\left\{X^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right)\right\} \\ &= P\left\{\bigcup_{i=1}^{\infty} X^{-1}(B_i)\right\} \\ &= \sum_{i=1}^{\infty} P\{X^{-1}(B_i)\} \\ &= \sum_{i=1}^{\infty} Q(B_i). \end{aligned}$$

It follows that $(\mathcal{R}, \mathcal{B}, Q)$ is a probability space.

Cumulative Distribution Function: Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$. Define a point function $F(\cdot)$ on $\mathbb{R}$, namely

$$F(x) = Q(-\infty, x] = P\{\omega : X(\omega) \leq x\} \quad \forall x \in \mathbb{R}.$$

The function F is called the *cumulative distribution function* of random variable X . We can simply write it as

$$F(x) = P\{X \leq x\}.$$

Properties of the Cumulative Distribution Function:

1. $0 \leq F(x) \leq 1$.
2. The distribution function F satisfies

$$F(-\infty) = \lim_{x \rightarrow -\infty} F(x) = 0 \text{ and } F(+\infty) = \lim_{x \rightarrow +\infty} F(x) = 1.$$

Proof: (i) Let $\{x_n\}$ be a decreasing sequence, such that, $\lim_{n \rightarrow \infty} x_n = -\infty$. Let

$$A_n = \{\omega : X(\omega) \leq x_n\}.$$

Then $\lim_{n \rightarrow \infty} A_n = \phi$. Now using continuity of probability measures

$$\begin{aligned} \lim_{n \rightarrow \infty} P(A_n) &= P(\lim_{n \rightarrow \infty} A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = P(\phi) = 0. \\ &\Rightarrow \lim_{n \rightarrow \infty} F(x_n) = 0. \\ &\Rightarrow \lim_{x \rightarrow -\infty} F(x) = 0. \end{aligned}$$

(ii) Let $\{x_n\}$ be an increasing sequence, such that, $\lim_{n \rightarrow \infty} x_n = +\infty$. Let

$$A_n = \{\omega : X(\omega) \leq x_n\}.$$

Then $\lim_{n \rightarrow \infty} A_n = \Omega$. Using continuity of probability measures

$$\begin{aligned} \lim_{n \rightarrow \infty} P(A_n) &= P(\lim_{n \rightarrow \infty} A_n) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = P(\Omega) = 1. \\ &\Rightarrow \lim_{n \rightarrow +\infty} F(x_n) = 1. \\ &\Rightarrow \lim_{x \rightarrow +\infty} F(x) = \bullet 1 \end{aligned}$$

3. F is nondecreasing, i.e., if $x < y$, then $F(x) \leq F(y)$.

Proof: If $x < y$, then

$$\begin{aligned} \{\omega : X(\omega) \leq x\} &\subseteq \{\omega : X(\omega) \leq y\} \\ \Rightarrow P\{\omega : X(\omega) \leq x\} &\leq P\{\omega : X(\omega) \leq y\}. \\ &\Rightarrow F(x) \leq F(y). \end{aligned}$$

4. F is right continuous.

Proof: Let $\{x_n\}$ be a decreasing sequence, such that, $\lim_{n \rightarrow \infty} x_n = x$. Let

$$A_n = \{\omega : X(\omega) \leq x_n\}.$$

Now

$$\begin{aligned} \lim_{n \rightarrow \infty} P(A_n) &= P(\lim_{n \rightarrow \infty} A_n) \\ \Rightarrow \lim_{n \rightarrow \infty} \{F(x_n) - F(x)\} &= 0. \\ \Rightarrow \lim_{n \rightarrow \infty} F(x_n) &= F(x). \end{aligned}$$

Since $\{x_n\}$ is a decreasing sequence to x (obviously from right), where $\{x_n\}$ is any arbitrary sequence, we can have

$$\Rightarrow \lim_{\epsilon \rightarrow 0} F(x + \epsilon) = F(x).$$

Remark 1: The aforementioned properties are satisfied by all cumulative distribution functions. Also, any function that satisfies these properties is necessarily a cumulative distribution function for some random variable X .

Theorem 2: The set of discontinuity points of a cumulative distribution function is at most countable.

Proof: Let a distribution function $F(x)$ has n points of discontinuities in a finite interval $[a, b]$, such that

$$A = \{x_1, x_2, \dots, x_n\}.$$

Since $A \subseteq [a, b]$,

$$P(A) \leq F(b) - F(a) \text{ or } \sum_{i=1}^n P(x_i) \leq F(b) - F(a).$$

and it follows that number of points x in $[a, b]$ with jump $p(x) > \epsilon > 0$ is at most $\epsilon^{-1} \{F(b) - F(a)\}$. Thus for every integer N , the number of discontinuity points with jump greater than $1/N$ is finite. It follows that there are no more than a countable number of discontinuity points in every finite interval $[a, b]$.

Discrete Random Variable: A random variable X defined on $(\Omega, \mathcal{F}, P)$ is said to be of discrete type, if there exists a countable set $E \subseteq \mathcal{R}$ such that $P(X \in E) = 1$. In simple terms, if a random variable X takes at most a countable number of values or countably infinite number of values, it

is called a *discrete random variable*. In other words, a real valued function defined on a discrete sample space is called a discrete random variable.

Probability Mass Function: Let X is a discrete random variable taking at most a countable infinite number of values $x_1, x_2, \dots$ with probability p_i ($i = 1, 2, \dots$). We have

$$P(X = x_i) = p_i, \quad i = 1, 2, \dots, \quad p_i \geq 0 \quad \forall \quad i.$$

Then $\sum_{i=1}^{\infty} p_i = 1$. The collection of numbers $\{p_i\}$ satisfying $P(X = x_i) = p_i \geq 0 \quad \forall \quad i$, and $\sum_{i=1}^{\infty} p_i = 1$, is called the *probability mass function* (PMF) of X .

The distribution function F of X with PMF, p_i is given by

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} p_i.$$

Result: $p_j = P(X = x_j) = F(x_j) - F(x_{j-1})$, $j = 1, 2, \dots$ where F is the distribution function of a random variable X .

Proof: Let $x_1 < x_2 < \dots$, we have

$$\begin{aligned} F(x_j) &= P(X \leq x_j) \\ &= \sum_{i=1}^j P(X = x_i) \\ &= \sum_{i=1}^j p_i, \end{aligned}$$

and

$$\begin{aligned} F(x_{j-1}) &= P(X \leq x_{j-1}) \\ &= \sum_{i=1}^{j-1} P(X = x_i) \\ &= \sum_{i=1}^{j-1} p_i. \end{aligned}$$

This implies that

$$F(x_j) - F(x_{j-1}) = p_j.$$



Example 1: The simplest example of a discrete random variable X is ‘ X degenerate at c ’, $P(X = c) = 1$. The corresponding CDF is given by

$$F(x) = \begin{cases} 0, & x < c, \\ 1, & x \geq c. \end{cases}$$

Example 2: A box contains good and defective items. If an item drawn is good, we assign the number 1 to the drawing; otherwise, the number 0. Let p be the probability of drawing a good item at random. Then $P(X = 0) = 1 - p$, $P(X = 1) = p$, and the corresponding CDF is given by

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & 1 \leq x. \end{cases}$$

Example 3: A random variable X has the following PMF:

x	0	1	2	3	4	5	6	7
$p(x)$	0	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2 + a$

Find the value of ‘ a ’ and then obtain the CDF of X .

Solution: Since $\sum_{x=0}^7 p(x) = 1$. One may easily obtain $a = 1/10$. The CDF is given by

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < 1, \\ 1/10, & 1 \leq x < 2, \\ 3/10, & 2 \leq x < 3, \\ 5/10, & 3 \leq x < 4, \\ 8/10, & 4 \leq x < 5, \\ 81/100, & 5 \leq x < 6, \\ 83/100, & 6 \leq x < 7, \\ 1, & x \geq 7. \end{cases}$$

Clearly, in this example, the graph of CDF will look like a *step function* having jump $p(x_i)$ at x_i ($i = 1, 2, \dots, 7$), and being constant between each pair of integers.

Continuous Random Variable and Probability Density Function: Let X be a random variable defined on $(\Omega, \mathcal{F}, P)$ with CDF F . Then X is said to be of continuous type if F is absolutely continuous, i.e., if there exists a nonnegative continuous function $f(x)$ such that for every real number x we have

$$F(x) = \int_{-\infty}^x f(t)dt.$$

The function f is called the *probability density function* (PDF) of the random variable X . The PDF f has the following properties:

1. $f \geq 0$.
2. $\lim_{x \rightarrow +\infty} F(x) = F(+\infty) = \int_{-\infty}^{\infty} f(t)dt = 1$.
3. Let a and b be any two real numbers with $a < b$, Then

$$P(a < X \leq b) = F(b) - F(a) = \int_a^b f(t)dt.$$

Remark 1: Let X be a continuous random variable with PDF f . Then for every Borel set $B \in \mathcal{B}$,

$$P(B) = \int_B f(t)dt.$$

Remark 2: If F is absolutely continuous and f is continuous at x , we have

$$F'(x) = \frac{dF(x)}{dx} = f(x).$$

Remark 3: Every nonnegative real function f that is integrable over $\mathbb{R}$ and satisfies

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

is the PDF of some continuous random variable X .

Remark 4: In the discrete case, $P(X = a)$ is the probability that X takes the value a . In the continuous case, $f(a)$ is not the probability that X takes the value a . Indeed, if X is of the continuous type, it assumes every value with probability 0.